



**GLOBAL ECONOMICS**  
**GRADE: 10TH**  
**CATCH-UP ACTIVITY**  
**ANALYSIS OF DECISIONS**  
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**FIRST**  
**TERM**  
**2025–2026**

**Learning objective:**

Formulate decision-making problems for optimal choice under uncertainty and evaluate alternatives using expected value.

**Assessment criteria:**

C1: Describes the way a problem can be formulated for optimal decision making. Evaluates expected value.

**Criteria assessment**

This activity evaluates criterion C1. Read each problem and complete the required decision-analysis tasks.

**1. Problem description**

A student entrepreneur is evaluating a short-term retail opportunity that depends heavily on how customer traffic develops during the weekend. The decision is straightforward in structure but uncertain in outcome, because actual demand can shift quickly based on consumer mood and local conditions. Management therefore needs a probability-based framework to connect each demand scenario with its corresponding financial consequence and to justify the final choice on expected performance rather than intuition.

*Payoff table*

| State of nature | Probability | Rent kiosk |
|-----------------|-------------|------------|
| High demand     | 0.60        | 36         |
| Low demand      | 0.40        | 10         |

Which alternative should management choose according to the Expected Value criterion?

**2. Problem description**

A coffee shop chain must select a bean purchasing approach for the next month while demand remains uncertain. One alternative emphasizes product quality and brand positioning, while the other emphasizes sourcing flexibility and cost control. Because the market can evolve in more than one direction, leadership must evaluate how each plan performs across possible demand states and use expected value reasoning to support a disciplined, forward-looking decision.

*Payoff table*

| State of nature | Probability | Plan A | Plan B |
|-----------------|-------------|--------|--------|
| High demand     | 0.55        | 84     | 72     |
| Low demand      | 0.45        | 22     | 34     |

Which alternative should management choose according to the Expected Value criterion?

**3. Problem description**

A small factory is planning seasonal production and must choose between two operating modes with different cost structures and responsiveness profiles. Market conditions may strengthen, remain stable, or soften, and each state creates a different profit implication for each mode. Since management cannot know in advance which condition will occur, the decision should be anchored in expected value so that uncertainty is incorporated systematically into the production strategy.

*Payoff table*

| State of nature | Probability | Mode A | Mode B |
|-----------------|-------------|--------|--------|
| Strong market   | 0.30        | 95     | 88     |
| Stable market   | 0.45        | 60     | 66     |
| Weak market     | 0.25        | 18     | 30     |

Which alternative should management choose according to the Expected Value criterion?

**4. Problem description**

A delivery startup is selecting a fleet strategy for the coming quarter in an environment where fuel conditions are uncertain and can materially affect operating margins. Each strategic option offers a different balance between control, flexibility, and exposure to cost volatility. To choose responsibly,

the firm must evaluate outcomes across the plausible fuel-cost states and rely on expected value to identify the alternative with the strongest overall economic justification.

| <i>Payoff table</i> |       |     |     |    |
|---------------------|-------|-----|-----|----|
| State of nature     | Prob. | A   | B   | C  |
| Low fuel cost       | 0.25  | 120 | 108 | 96 |
| Medium fuel cost    | 0.50  | 86  | 92  | 88 |
| High fuel cost      | 0.25  | 44  | 58  | 70 |

Which alternative should management choose according to the Expected Value criterion?

### 5. Problem description

A supermarket is defining an inventory policy for imported fruit while facing uncertainty in supply-chain reliability. Management recognizes that logistics conditions can range from smooth to severely disrupted, and each situation affects availability, waste risk, and revenue potential differently. Because no single scenario is guaranteed, the firm must compare policies using probability-weighted outcomes and select the option that best supports resilient profit performance.

| <i>Payoff table</i>       |       |    |    |    |
|---------------------------|-------|----|----|----|
| State of nature           | Prob. | A  | B  | C  |
| Very smooth supply        | 0.20  | 74 | 82 | 86 |
| Smooth supply             | 0.35  | 68 | 76 | 78 |
| Disrupted supply          | 0.30  | 36 | 50 | 58 |
| Severely disrupted supply | 0.15  | 8  | 24 | 34 |

Which alternative should management choose according to the Expected Value criterion?

### 6. Problem description

An electronics retailer is choosing a launch format for a new device in a season where demand intensity may vary significantly. Each format reflects a different channel strategy and operational commitment, leading to distinct financial outcomes under different market conditions. Decision-makers therefore need to assess each format across plausible seasonal states and use expected value analysis to justify the launch approach with the strongest overall return profile.

*Payoff table*

| State of nature | Prob. | A   | B   | C   | D   |
|-----------------|-------|-----|-----|-----|-----|
| Boom season     | 0.20  | 150 | 132 | 144 | 120 |
| Good season     | 0.30  | 110 | 118 | 124 | 108 |
| Moderate season | 0.30  | 72  | 82  | 94  | 88  |
| Slow season     | 0.20  | 30  | 48  | 62  | 66  |

Which alternative should management choose according to the Expected Value criterion?

### 7. Problem description

A grain exporter must choose among competing shipping contracts while freight market conditions remain uncertain. Contract design influences both upside potential and downside protection as freight tightness changes across the planning horizon. Since management cannot predict a single freight outcome with certainty, it should evaluate all contracts under the full set of plausible states and apply expected value logic to support a defensible commercial decision.

| <i>Payoff table</i>    |       |     |     |     |     |
|------------------------|-------|-----|-----|-----|-----|
| State of nature        | Prob. | A   | B   | C   | D   |
| Very favorable freight | 0.15  | 160 | 152 | 146 | 132 |
| Favorable freight      | 0.25  | 138 | 140 | 136 | 126 |
| Neutral freight        | 0.30  | 102 | 114 | 120 | 116 |
| Tight freight          | 0.20  | 66  | 80  | 92  | 98  |
| Very tight freight     | 0.10  | 24  | 40  | 54  | 70  |

Which alternative should management choose according to the Expected Value criterion?

### 8. Problem description

A fashion retailer is evaluating expansion plans for urban markets where demand can shift across several intensity levels. Each plan reflects a different growth posture, creating distinct exposure to both strong and weak market outcomes. To align strategy with financial discipline, management should compare alternatives across all plausible demand states and use expected value as the core criterion for selecting the expansion path.

| <i>Payoff table</i> |       |     |     |     |     |     |
|---------------------|-------|-----|-----|-----|-----|-----|
| State               | Prob. | A   | B   | C   | D   | E   |
| Very high           | 0.12  | 190 | 182 | 176 | 168 | 156 |
| High                | 0.23  | 150 | 154 | 152 | 146 | 140 |
| Medium              | 0.30  | 108 | 120 | 128 | 126 | 124 |
| Low                 | 0.22  | 58  | 76  | 90  | 98  | 104 |
| Very low demand     | 0.13  | 12  | 34  | 52  | 64  | 76  |

Which alternative should management choose according to the Expected Value criterion?

**9. Problem description**

A regional energy distributor is selecting a pricing package in a context where weather-driven demand can change materially during the operating period. Each package offers a different trade-off between high-demand capture and low-demand protection. Because weather patterns are uncertain and financially significant, management needs a probability-weighted evaluation of outcomes to determine which package provides the most robust expected economic result.

| Payoff table |       |     |     |     |     |     |
|--------------|-------|-----|-----|-----|-----|-----|
| State        | Prob. | A   | B   | C   | D   | E   |
| Extreme cold | 0.10  | 220 | 210 | 204 | 196 | 186 |
| Cold         | 0.20  | 184 | 186 | 182 | 176 | 170 |
| Mild         | 0.25  | 142 | 152 | 158 | 156 | 152 |
| Warm         | 0.20  | 96  | 110 | 122 | 128 | 130 |
| Hot          | 0.15  | 58  | 74  | 88  | 98  | 106 |
| Extreme hot  | 0.10  | 22  | 40  | 54  | 68  | 82  |

Which alternative should management choose according to the Expected Value criterion?

**10. Problem description**

A national logistics group is deciding on a network design while macro-economic conditions may evolve through expansion, slowdown, contraction, and recovery patterns. Each design presents a different balance between growth capacity and resilience under weaker environments. Given this broad uncertainty set, executives should compare all designs across the plausible states of nature and rely on expected value analysis to choose the configuration with the strongest expected performance.

| Payoff table       |       |     |     |     |     |     |
|--------------------|-------|-----|-----|-----|-----|-----|
| State of nature    | Prob. | A   | B   | C   | D   | E   |
| Rapid expansion    | 0.14  | 260 | 248 | 242 | 234 | 226 |
| Steady growth      | 0.20  | 220 | 222 | 218 | 212 | 206 |
| Flat market        | 0.24  | 170 | 182 | 188 | 190 | 188 |
| Mild contraction   | 0.16  | 118 | 132 | 146 | 156 | 162 |
| Strong contraction | 0.12  | 70  | 86  | 102 | 118 | 130 |
| Recovery           | 0.14  | 136 | 148 | 158 | 164 | 168 |

Which alternative should management choose according to the Expected Value criterion?